Tool	Layer	Description and enterprise use	Maturity
Kubernetes	**Orchestration**	The *de facto* container orchestration platform. Manages scheduling, scaling, and fault tolerance for all AI workload types. Foundation layer for all cloud-native AI infrastructure.	**Production**
Kubeflow	**ML platform**	End-to-end ML orchestration on Kubernetes. Provides pipelines, notebooks, KServe for model serving, and training operators for distributed jobs. Originally developed by Google, it is now CNCF-hosted.	**Production**
MLflow	**Experiment tracking**	Lifecycle management for ML models: experiment tracking, model registry, packaging, and deployment. Framework-agnostic and cloud-agnostic. Widely adopted as the standard for model reproducibility and collaborative development.	**Production**
Ray	**Distributed compute**	Unified framework for scaling Python and AI workloads from laptop to cluster. Ray Train handles distributed training; Ray Serve handles scalable model serving; Ray Tune handles hyperparameter optimisation. Particularly strong for LLM fine-tuning.	**Production**
Apache Airflow	**Pipeline orchestration**	DAG-based workflow orchestration for data and ML pipelines. Manages scheduling, dependencies, retries, and alerting across complex multi-step pipelines. Broadly adopted for production data engineering and ML batch workflows.	**Production**
KServe (formerly KFServing)	**Model serving**	Kubernetes-native model inference platform supporting multi-framework serving (TensorFlow, PyTorch, ONNX, Triton). Provides auto-scaling, canary deployments, and explainability. The standard for a scalable, production model serving on Kubernetes.	**Production**
NVIDIA Triton Inference Server	**Model serving**	High-performance inference server supporting multiple frameworks and hardware backends. Features dynamic batching, model ensembling, and GPU-optimised execution. Best-in-class throughput for latency-sensitive inference workloads.	**Production**
Feast	**Feature store**	Operational feature store for managing, storing, and serving ML features. Ensures consistency between training-time and serving-time feature computation. Eliminates training-serving skew — a critical production reliability concern.	**Production**
DVC (Data Version Control)	**Data versioning**	Git-based version control for datasets, models, and ML experiments. Enables reproducibility by tracking data and model artifacts alongside code. Integrates with major cloud storage backends and CI/CD pipelines.	**Production**
Seldon Core	**Model serving**	Kubernetes-native framework for deploying, scaling, and monitoring ML models. Supports advanced inference graphs (transformers, routers, combiners) and A/B testing. Cloud-agnostic with strong enterprise governance features.	**Production**
Prometheus + Grafana	**Observability**	The standard open source observability stack. Prometheus for metrics collection and alerting; Grafana for visualisation and dashboarding. Extended with AI-specific exporters for GPU metrics (DCGM Exporter) and model performance.	**Production**
Apache Kafka	**Data streaming**	Distributed event streaming platform for high-throughput, fault-tolerant data pipelines. Foundational for real-time feature computation, online learning pipelines, and streaming inference workloads in production AI systems.	**Production**
Apache Spark	**Data processing**	Unified analytics engine for large-scale data processing. Critical for feature engineering at scale, training data preparation, and batch inference on large datasets. Integrates with Kubernetes via the Spark Operator.	**Production**
Horovod	**Distributed training**	Distributed deep learning training framework for TensorFlow, Keras, and PyTorch. Implements the Ring-AllReduce algorithm for efficient gradient synchronisation across large GPU clusters, significantly reducing training time at scale.	**Production**
Flyte	**Workflow orchestration**	Cloud-native platform for orchestrating ML and data workflows with strong typing, reproducibility, and scheduling. Originally built at Lyft. Preferred for teams requiring strict type safety and large-scale production pipeline management.	**Production**
MinIO	**Object storage**	High-performance, S3-compatible object storage. The standard for on-premises or private cloud AI data storage — model artifacts, training datasets, checkpoints, and pipeline outputs. Kubernetes-native and exceptionally performant at scale.	**Production**
LangChain / LlamaIndex	**LLM orchestration**	Frameworks for building LLM-powered applications with retrieval augmentation, agent workflows, and external data integration. Increasingly central to enterprise genAI application infrastructure built on top of foundation model serving layers.	**Maturing**
BentoML	**Model serving**	Open platform for packaging, serving, and deploying ML models across frameworks. Simplifies the operational gap between model development and production serving with a unified artifact format and deployment abstraction.	**Maturing**